

Chapter 10: Mechanics of Options Markets

Part 3 — Sections 10.7 to 10.9

Learning Objectives

After studying this part, you should be able to:

- Understand margin requirements in options markets.
 - Calculate margin requirements for naked call and put positions.
 - Understand how covered calls affect margin requirements.
 - Understand the role of the Options Clearing Corporation (OCC).
 - Understand the option exercise and assignment process.
 - Understand position limits and exercise limits.
 - Understand the purpose of regulations in options markets.
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10.7 Margin Requirements

Unlike option buyers, option writers face potentially large future obligations.

Because of this:

Option writers are required to maintain margin accounts.

The purpose of margin is to ensure that option writers can fulfill their contractual obligations if options are exercised.

Margin for Option Buyers

Short-Term Options

When purchasing options with maturities less than 9 months:

- Premium must be paid in full.
- No borrowing is allowed.

Reason:

Options already provide significant leverage.

Allowing additional borrowing would create excessive risk.

Long-Term Options

For options with maturities greater than 9 months:

Investors may borrow up to:

25%

of the option value.

Why Writers Must Post Margin

An option writer:

- Receives premium immediately.
- May face substantial future losses.

The exchange therefore requires collateral.

Naked Options

A naked option is:

An option written without an offsetting position in the underlying asset.

Naked Call

Example:

Investor writes a call but does not own the stock.

If stock price rises sharply:

Losses can become extremely large.

Naked Put

Example:

Investor writes a put without any offsetting hedge.

If stock price collapses:

Large losses may occur.

Margin Requirement for Naked Calls

The CBOE requires the greater of:

Calculation 1

$$\text{Option Premium} + 20\% \times S - \text{Out-of-Money Amount}$$

Calculation 2

$$\text{Option Premium} + 10\% \times S$$

The larger amount becomes the required margin.

Margin Requirement for Naked Puts

The greater of:

Calculation 1

$$\text{Option Premium} + 20\% \times S - \text{Out-of-Money Amount}$$

Calculation 2

$$\text{Option Premium} + 10\% \times K$$

where:

$$K = \text{Strike Price}$$

Index Option Exception

For broadly diversified stock indices:

The 20% figure is often reduced to:

15%

because indices are generally less volatile than individual stocks.

Example: Margin on Naked Call

Suppose:

- Four call contracts written
- Option premium = \$5
- Strike price = \$40
- Stock price = \$38

Each contract covers:

100 shares

Total shares:

400

Step 1: Determine Moneyness

Option is:

$$40 - 38 = 2$$

out of the money.

Step 2: First Calculation

$$\begin{aligned} &400(5 + 0.2(38) - 2) \\ &= 400(10.6) \\ &= 4240 \end{aligned}$$

Step 3: Second Calculation

$$\begin{aligned} &400(5 + 0.1(38)) \\ &= 400(8.8) \\ &= 3520 \end{aligned}$$

Required Margin

Greater amount:

4240

Therefore:

$\text{Margin Requirement} = 4240$

Daily Margin Adjustments

Margin calculations are repeated every day.

If Margin Requirement Falls

Investor may withdraw funds.

If Margin Requirement Rises

Broker issues a:

Margin Call

Investor must deposit additional funds.

Covered Calls

A covered call occurs when:

- Investor owns the underlying shares.
 - Investor writes a call option on those shares.
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Why Covered Calls Are Safer

If assigned:

Shares are already owned.

No need to purchase shares at potentially high market prices.

Margin Treatment

Covered calls require substantially less margin than naked calls.

This reflects their lower risk.

Why Margin Matters

Margin systems:

- Reduce default risk.
 - Protect exchanges.
 - Protect counterparties.
 - Improve market stability.
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10.8 The Options Clearing Corporation (OCC)

The OCC performs for options markets a role similar to that of clearing houses in futures markets.

Main Functions of the OCC

The OCC:

- Guarantees contract performance.
 - Maintains records of positions.
 - Handles exercises and assignments.
 - Reduces counterparty risk.
 - Monitors margin requirements.
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Importance of the OCC

Without the OCC:

Option holders would need to worry about whether writers could pay.

With OCC guarantees:

Participants focus only on market risk.

Credit risk is greatly reduced.

Membership Requirements

OCC members must:

- Maintain minimum capital.
- Meet regulatory standards.
- Contribute to special protection funds.

These requirements help protect the market against defaults.

Flow of Funds

Consider an option purchase.

Step 1

Investor purchases option.

Step 2

Broker receives funds.

Step 3

Broker clears through an OCC member.

Step 4

OCC records the position.

This creates a centralized clearing system.

Exercising an Option

When an investor chooses to exercise:

The process follows a defined sequence.

Step 1

Investor instructs broker.

Step 2

Broker notifies OCC member.

Step 3

OCC member submits exercise notice to OCC.

Step 4

OCC randomly selects a member with an outstanding short position.

Step 5

That member assigns the exercise to one of its customers with a short position.

Assignment

Assignment occurs when a short option writer is selected to fulfill an exercised option.

Call Assignment

Writer must:

Sell shares at strike price.

Put Assignment

Writer must:

Purchase shares at strike price.

Significance of Assignment Risk

Short option positions carry:

Assignment Risk

The writer never knows exactly when assignment may occur.

This is particularly important for American options because they can be exercised before expiration.

Benefits of Centralized Clearing

The OCC:

Reduces Credit Risk

Counterparty default risk is minimized.

Improves Liquidity

Investors can trade without worrying about counterparty quality.

Standardizes Operations

All exercises and assignments follow uniform procedures.

10.9 Regulation

Options markets require regulations to ensure orderly trading.

One important regulatory tool is the use of position limits.

Position Limits

Position limits restrict:

The maximum number of contracts an investor may hold.

Why Position Limits Exist

Without limits:

Large traders could potentially influence prices.

Position limits help prevent market manipulation.

Typical Limits

For highly liquid stocks:

250,000

contracts may be allowed.

Smaller capitalization stocks may have limits of:

- 200,000
- 75,000
- 50,000
- 25,000

contracts.

Exercise Limits

Exercise limits restrict:

The maximum number of contracts that can be exercised within a specified period.

Purpose of Exercise Limits

These limits prevent:

- Market distortions
 - Excessive concentration
 - Manipulative trading strategies
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Example

Suppose a trader accumulates a massive number of call options.

Without exercise limits:

The trader might attempt to exercise all contracts simultaneously and create market disruption.

Exercise limits reduce this possibility.

Are Limits Always Necessary?

Hull notes that the usefulness of position and exercise limits remains controversial.

Arguments Supporting Limits

- Reduce manipulation.
 - Improve market stability.
 - Protect smaller participants.
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Arguments Against Limits

- Reduce market efficiency.
 - Restrict legitimate hedging.
 - Limit institutional trading flexibility.
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Regulatory Goals

The overall goals of option market regulation are:

Fairness

Equal treatment of participants.

Transparency

Clear market information.

Stability

Reduced risk of market disruptions.

Integrity

Prevention of manipulation.

Relationship Between Regulation and Liquidity

Well-designed regulation can:

- Increase confidence.
- Attract participants.
- Improve liquidity.

Poorly designed regulation can:

- Reduce participation.
- Increase trading costs.

Section Summary

Margin Requirements

Required primarily for option writers.

Purpose:

Protect against default.

Naked Option

Written without an offsetting position.

Higher risk.

Higher margin requirement.

Covered Call

Written against owned stock.

Lower risk.

Lower margin requirement.

OCC

Responsible for:

- Clearing
 - Settlement
 - Assignment
 - Performance guarantees
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Assignment

Occurs when a short option writer is selected to fulfill an exercised option.

Position Limits

Restrict maximum contracts held.

Exercise Limits

Restrict maximum contracts exercised.

Regulatory Objective

Maintain:

- Fairness
 - Stability
 - Market integrity
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Quick Revision Sheet

Naked Option

Unhedged written option.

Covered Call

Written call backed by owned shares.

Margin for Naked Calls

Greater of:

$$Premium + 20\%S - OTM$$

or

$$Premium + 10\%S$$

Margin for Naked Puts

Greater of:

$$Premium + 20\%S - OTM$$

or

$$Premium + 10\%K$$

OCC Functions

- Guarantees performance
 - Clears trades
 - Handles assignments
 - Maintains records
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Assignment

Short writer selected to fulfill exercised option.

Position Limits

Maximum contracts held.

Exercise Limits

Maximum contracts exercised.

Main Purpose of Regulation

Prevent manipulation and maintain orderly markets.

End of Part 3 (Sections 10.7–10.9)